Arduino-Based Line Following Robot

Autonomous Robot Using Arduino Uno

Arduino Uno | **Line Following** | **Obstacle Detection**

Course : Prototype & System Engineering
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Team Members:

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4. Aynul Haque

Project Objectives

Build a fully autonomous robot that follows lines, detects obstacles, and responds safely — from design through demonstration.

Design & Build

Design and build an autonomous line-following robot

Line Sensing

Follow a black line using IR sensors

Obstacle Detection

Detect obstacles using ultrasonic sensors

Safe Response

Stop and respond safely when an obstacle is detected

Full Pipeline

Build the full system: design → simulation → wiring → programming

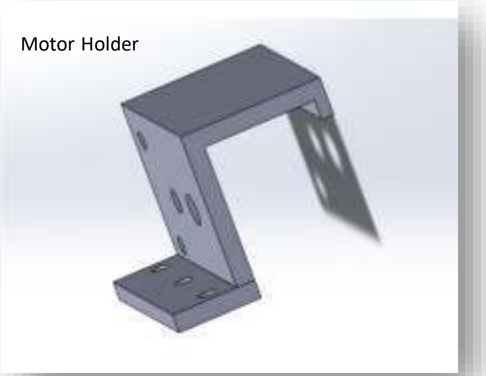
Robot Design and Holders

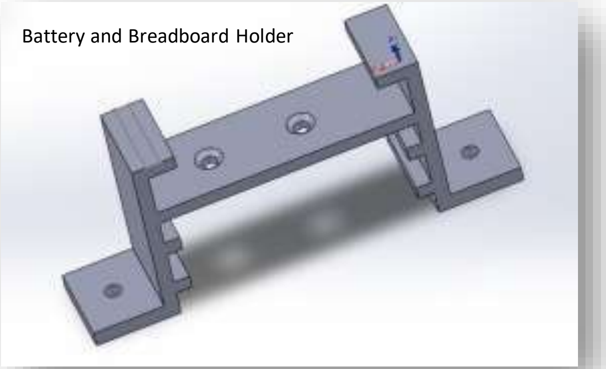
A stable, well-mounted chassis forms the mechanical foundation — no glue, tape, or zip ties.

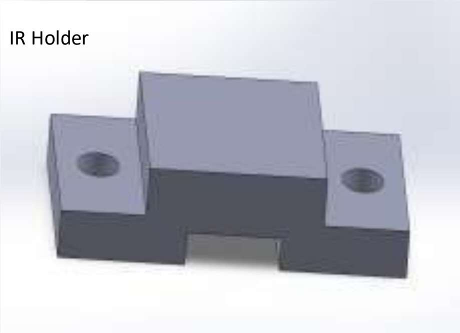
Mechanical Design

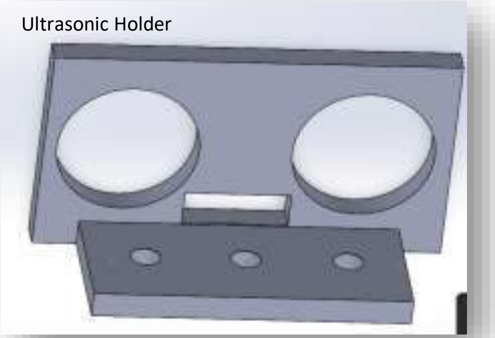
- ✓ Designed the robot chassis and component layout
- ✓ Mounted the Arduino and motor driver securely
- ✓ Used holders for IR and ultrasonic sensors
- ✓ Kept design stable for real movement and testing
- ✓ Avoided glue, tape, and zip ties

Sensor Holder Bracket — CAD Render







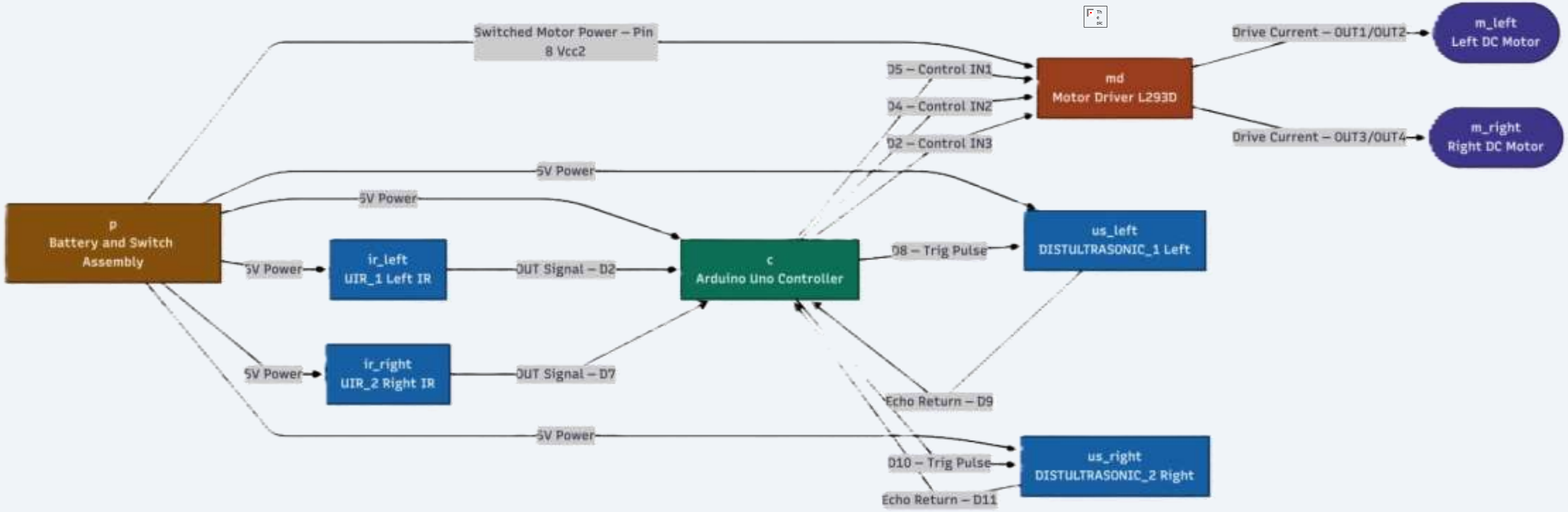


Circuit Diagram & Connection Plan

A complete circuit diagram was planned before any hardware assembly — ensuring correct pin assignments and a clean common ground.

- Created the full circuit diagram before hardware assembly
- Assigned Arduino pins for motors, sensors, and power
- Made the circuit easy to follow for simulation and hardware implementation

Connection Flow



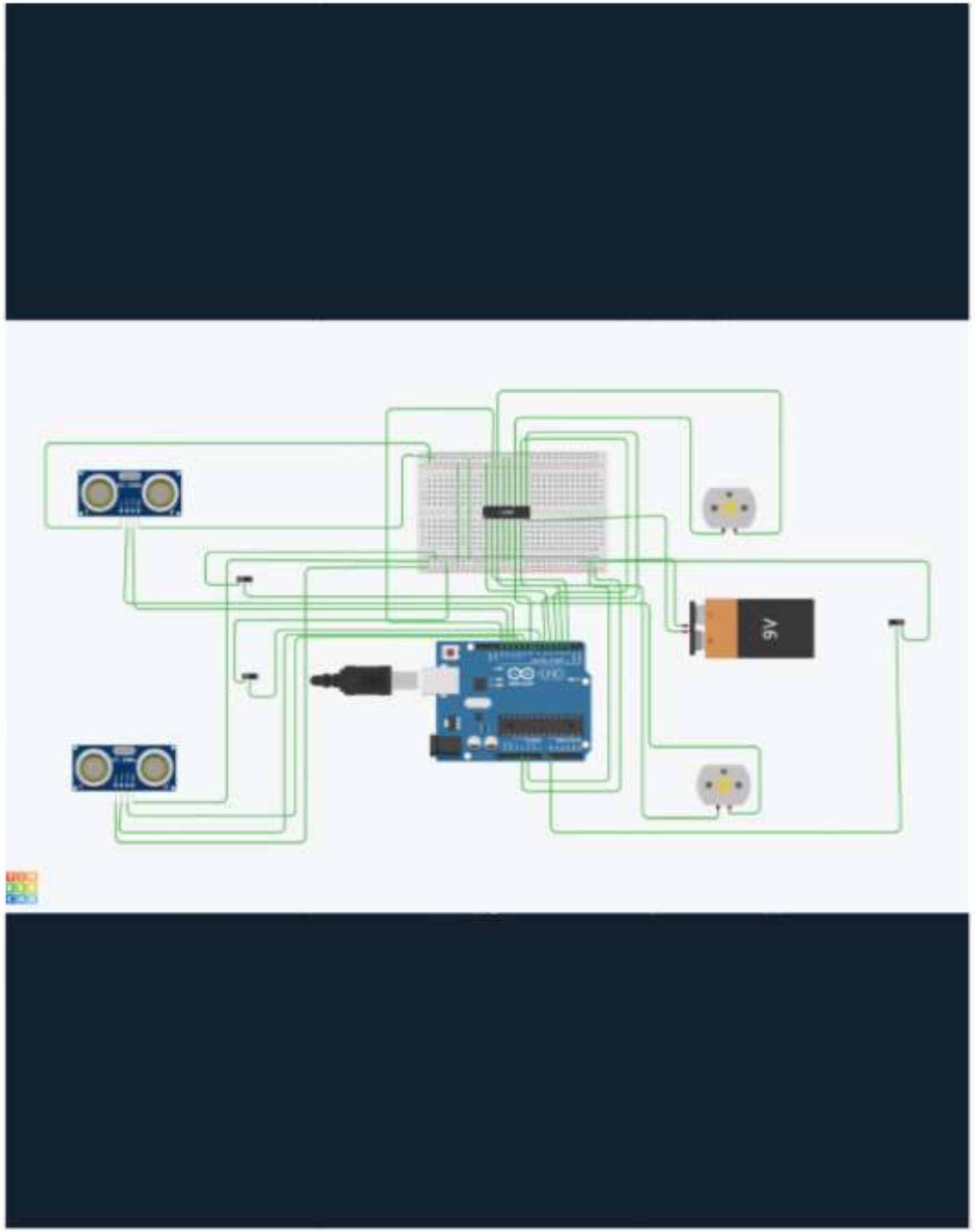
Tinkercad Simulation

Virtual simulation validated wiring logic and robot behavior before a single physical component was connected.

- Recreated the full circuit in Tinkercad
- Checked wiring logic and Arduino pin assignments
- Used switches to represent IR line sensors in simulation
- Verified robot behavior before using real hardware
- Visualized the whole project before physical assembly

Key Benefit: Simulation reduced hardware mistakes

Tinkercad — Circuit Simulation

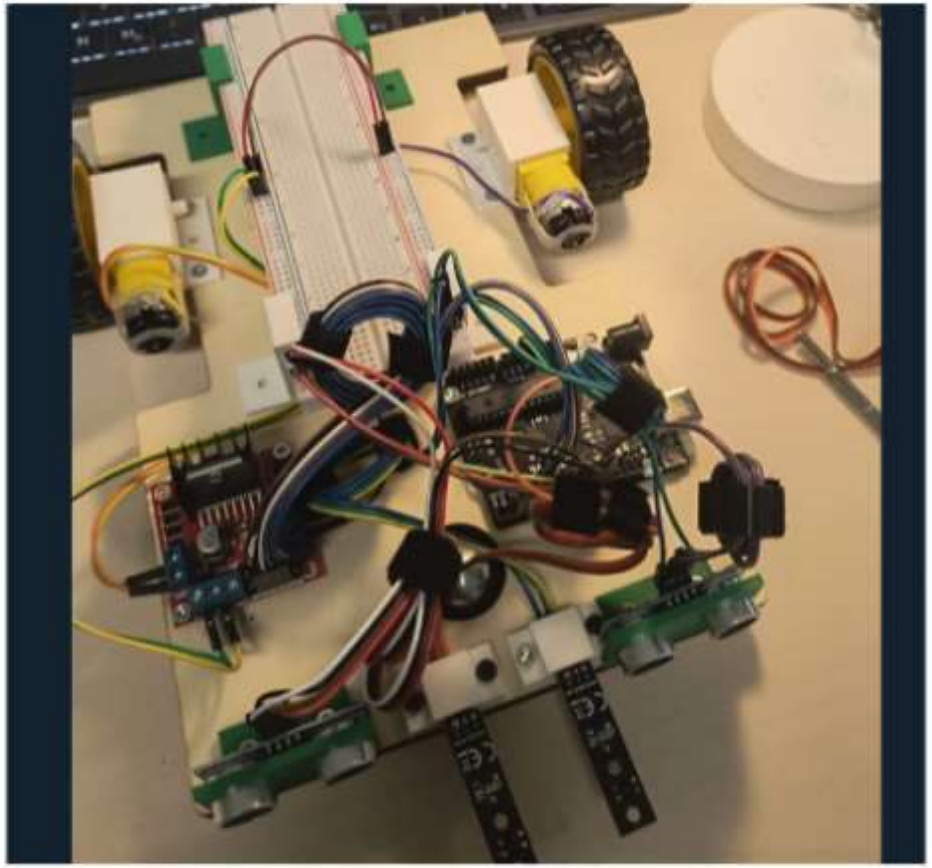


Physical Wiring & Assembly

The circuit diagram was translated into a complete physical build with all modules properly wired and grounded.

Hardware Implementation

- Converted the diagram into a real physical build
- Connected breadboard, motor driver, motors, sensors, and battery
- Wired IR sensors and ultrasonic sensors to assigned pins
- Removed ENA and ENB jumpers for PWM speed control
- Connected a common ground between all modules



Demonstration

The robot successfully performs three behaviors: line following, obstacle avoidance, and 180-degree U-turn.

1

IR Sensors

Line Following

Robot follows the black path, continuously reading IR sensor signals to stay on course and correct its direction in real time.

2

Ultrasonic Sensor

Obstacle Avoidance

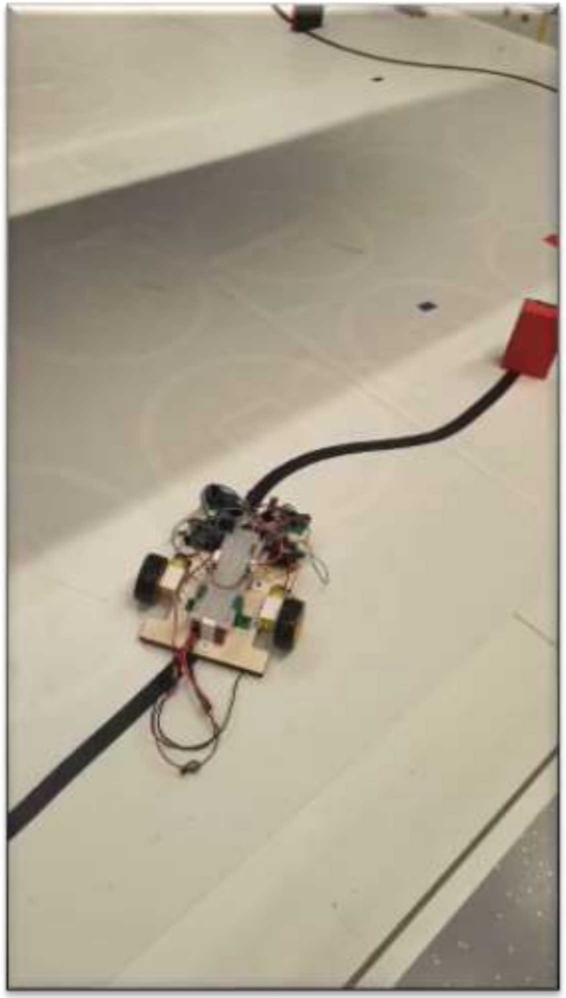
Robot detects an obstacle in its path using ultrasonic distance sensing and brings itself to a full stop to avoid collision.

3

180° U-Turn Behavior

Robot changes direction after detecting an obstacle or losing the path, executing a full 180-degree turn to resume operation.

Motor Control



Challenges & Future Improvements

Real-world challenges sharpened the team's skills, and clear improvements are identified for the next iteration.

Challenges Faced

- Correct L298N motor driver wiring
- Designing the Holders
- Tackling the Obstacle Avoidance
- 180 Degree U turn
- Right Placement of the Sensors

Future Improvements

- PID control for smoother line following
- Industrial Application
- Bluetooth or wireless control
- Autonomous Path Planning
- Maze solving and smarter navigation

Final Slide

Conclusion

Through teamwork, simulation, and iterative testing, the team successfully delivered a fully functional autonomous robot.

Thank You

Questions?

Key Achievements

Design, Simulation, Wiring & Programming
Successfully completed the full development lifecycle from concept to working hardware

Unified Line Following & Obstacle Detection
Combined IR sensor-based line tracking and ultrasonic obstacle avoidance in one system

Teamwork & Systematic Testing
Delivered the project through collaborative effort and structured iterative testing

Design → Simulation → Wiring → Programming → Demo